

UNIT III

Sequential Logic And Its Applications: Storage elements: latches & flip flops, Characteristic Equations of Flip Flops, Flip Flop Conversion, Shift Registers, Ripple Counters, Synchronous Counters, Other Counters: Johnson & Ring Counter.

1 What is Sequential Logic?

In combinational circuits:

Output depends only on present input.

In sequential circuits:

Output depends on:

- Present input
- Past output (memory)

Mathematically:

Output = $f(\text{Present Input, Past State})$

Why Memory is Needed?

Example:

Counter.

If you want to count:

$0 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 4$

The system must remember previous number.

That memory is stored using **latches and flip-flops**.

2 Storage Elements

Storage elements store one bit (0 or 1).

Two main types:

1. Latches
2. Flip-Flops

3 Latches

What is a Latch?

A latch is a basic memory element.

It is:

- Level triggered
- Output changes as long as input is active

3.1 SR Latch (Set-Reset Latch)

Inputs:

S (Set)

R (Reset)

Outputs:

Q

Q'

Truth Table

S	R	Q (Next State)
0	0	No change
0	1	0
1	0	1
1	1	Invalid

Problem

S=1 and R=1 → invalid state.

3.2 D Latch

To remove invalid condition.

D latch has:

- One input D
- One Enable (E)

Q follows D when enable is 1.

Equation:

Q = D]

Used in:

- Simple memory circuits

4 Flip-Flops

Flip-flops are improved latches.

Difference:

Flip-flop is **edge-triggered** (works only on clock edge).

Clock signal controls operation.

Types of Flip-Flops

1. SR Flip-Flop
2. JK Flip-Flop
3. D Flip-Flop
4. T Flip-Flop

4.1 SR Flip-Flop

Same as SR latch but clock controlled.
Invalid state still exists.

4.2 JK Flip-Flop

Improved SR flip-flop.
No invalid condition.

Truth Table

J	K	Q(next)
0	0	No change
0	1	0
1	0	1
1	1	Toggle

Toggle means:

If $Q=0 \rightarrow$ becomes 1

If $Q=1 \rightarrow$ becomes 0

Characteristic Equation

$$Q(\text{next}) = JQ' + K'Q$$

4.3 D Flip-Flop

Most commonly used.

Input:

D

Output:

$$Q(\text{next}) = D$$

Very simple.

Used in:

- Registers
- Memory
- Pipeline stages in CPU

4.4 T Flip-Flop

T stands for Toggle.

If $T=1 \rightarrow$ toggle

If $T=0 \rightarrow$ no change

Equation:

$$Q(\text{next}) = T \oplus Q$$

Used in:

- Counters

5 Characteristic Equations of Flip-Flops

Flip-Flop	Characteristic Equation
SR	$Q(\text{next}) = S + R'Q$
JK	$Q(\text{next}) = JQ' + K'Q$
D	$Q(\text{next}) = D$
T	$Q(\text{next}) = T \oplus Q$

Very important for exam derivation.

6 Flip-Flop Conversion

Sometimes we need to convert:

- $JK \rightarrow D$
- $D \rightarrow T$
- $SR \rightarrow JK$

Method:

1. Write excitation table
2. Compare characteristic equations
3. Derive required input expression

Example:

Convert JK to D.

We know:

$$D \text{ flip-flop} \rightarrow Q(\text{next}) = D$$

$$JK \rightarrow Q(\text{next}) = JQ' + K'Q$$

So equate both and find J and K values.

7 Shift Registers

A shift register is a group of flip-flops connected in series.

Used to store and shift data.

Types of Shift Registers

1. SISO (Serial In Serial Out)
2. SIPO (Serial In Parallel Out)
3. PISO (Parallel In Serial Out)
4. PIPO (Parallel In Parallel Out)

Example: SISO

Data enters one bit at a time.

After each clock pulse → data shifts right.

Used in:

- Data transmission
- Communication systems

8 Counters

Counter counts clock pulses.

It is made using flip-flops.

8.1 Ripple Counter (Asynchronous Counter)

Clock given only to first flip-flop.

Next flip-flop triggered by previous output.

Features

- Simple design
- Delay accumulates
- Slow for high frequency

Example:

Binary counter 0 → 1 → 2 → 3 → 4

8.2 Synchronous Counter

All flip-flops receive clock at same time.

Advantages

- Fast
- No ripple delay
- Used in processors

9 Ring Counter

Made using shift register.

Last output connected to first input.

If 4 flip-flops → 4 states only.

Example:

1000 → 0100 → 0010 → 0001 → repeat

Used in:

- Sequence generation
- Control circuits

10 Johnson Counter

Modified ring counter.

Inverted output of last flip-flop fed to first.

If n flip-flops → $2n$ states.

Example (4 FF):

0000 → 1000 → 1100 → 1110 → 1111 → 0111 → 0011 → 0001 → repeat

More states than ring counter.

Used in:

- Timing circuits
- Sequence generators

Ripple vs Synchronous Counter

Ripple Counter	Synchronous Counter
Clock to first FF only	Clock to all FF
Slow	Fast
Simple design	Complex design
Ripple delay present	No ripple delay

1 Latch vs Flip-Flop

Feature	Latch	Flip-Flop
Trigger Type	Level triggered	Edge triggered
Clock Required	No (enable based)	Yes (clock pulse)
Control	Changes while enable is active	Changes only at clock edge
Speed	Faster (simple)	Slightly slower
Reliability	Less stable	More stable
Used In	Simple storage	Registers, counters

★ Important exam line:

Latch → Level sensitive

Flip-flop → Edge sensitive

2 SR Flip-Flop vs JK Flip-Flop

Feature	SR Flip-Flop	JK Flip-Flop
Inputs	S, R	J, K
Invalid Condition	Yes (S=R=1)	No
Toggle Operation	Not possible	Yes (J=K=1)
Stability	Less	More stable
Used In	Basic circuits	Counters

3 JK Flip-Flop vs D Flip-Flop

Feature	JK Flip-Flop	D Flip-Flop
Inputs	J, K	D
Toggle Mode	Yes	No
Characteristic Equation	$Q^+ = JQ' + K'Q$	$Q^+ = D$
Complexity	More logic	Simple
Used In	Counters	Registers, memory

4 D Flip-Flop vs T Flip-Flop

Feature	D Flip-Flop	T Flip-Flop
Input	D	T
Operation	$Q^+ = D$	$Q^+ = T \oplus Q$
Toggle	No	Yes when T=1

Used In	Data storage	Counters
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5 SR Latch vs SR Flip-Flop

Feature	SR Latch	SR Flip-Flop
Trigger	Level	Edge
Clock	No	Yes
Stability	Less	More
Application	Temporary storage	Sequential circuits

6 Shift Register vs Counter

Feature	Shift Register	Counter
Purpose	Store & shift data	Count clock pulses
Operation	Shift bits left/right	Increment/decrement
Built Using	Flip-flops	Flip-flops
Used In	Communication	Timers, clocks

7 Ripple Counter vs Synchronous Counter

Feature	Ripple Counter	Synchronous Counter
Clock Input	First FF only	All FFs
Speed	Slow	Fast
Delay	Ripple delay present	No ripple delay
Design	Simple	Complex
Used In	Low-speed systems	Processors

★ Very important difference question.

8 Ring Counter vs Johnson Counter

Feature	Ring Counter	Johnson Counter
Feedback	Direct output fed back	Inverted output fed back
States	n states (for n FF)	2n states
Efficiency	Less	More
Initial State	One bit = 1	All zeros allowed
Used In	Sequence generator	Timing circuits

9 Serial vs Parallel Data Transfer (Shift Register Context)

Feature	Serial	Parallel
Bits Transferred	One at a time	All at once
Speed	Slow	Fast
Wires Needed	Fewer	More
Used In	Communication	CPU internal data

10 Characteristic Equation vs Excitation Table

Feature	Characteristic Equation	Excitation Table
Purpose	Gives next state	Gives required inputs
Used For	State analysis	Flip-flop conversion
Based On	Present state + input	Present state + next state

[GKM](#)